Certainty Method	Llama	Mistral	Gemma	Llama-Inst	Mistral-Inst	Gemma-Inst
Probability	$11.0_{\pm 1.3}$	$17.6_{\pm 2.0}$	$10.3_{\pm 3.3}$	$27.4_{\pm 3.1}$	$22.4_{\pm 3.1}$	$24.8_{\pm 9.6}$
Probability Diff.	$9.2_{\pm 2.6}$	$18.1_{\pm 2.8}$	$10.6_{\pm 5.1}$	$25.3_{\pm 4.8}$	$21.8_{\pm 2.2}$	$22.9_{\pm 7.2}$
Semantic Entropy	$11.1_{\pm 1.8}$	$11.2_{\pm 2.3}$	$12.6_{\pm 2.3}$	$14.9_{\pm 3.4}$	$22.8_{\pm 3.4}$	$20.7_{\pm 3.6}$
Metrics Intersection	$4.04_{\pm 1.57}$	$4.37_{\pm 1.39}$	$2.77_{\pm 0.60}$	$7.01_{\pm 1.24}$	$9.82_{\pm 2.14}$	$11.15_{\pm 1.65}$

Table 4: We show that across models and certainty methods, CHOKE examples occur at average rates of 8–26% on TriviaQA. **Key finding:** A substantial portion of hallucinations persist at high certainty levels, demonstrating that models can produce certain hallucinations even when they possess the correct information.

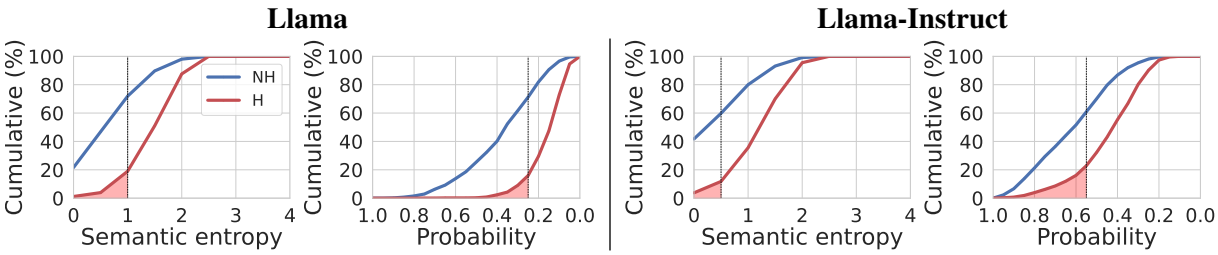

Figure 7: **Analysis of CHOKE across models and certainty metrics.** Cumulative distributions of hallucinations (H) and correct answers (NH) when models possess correct knowledge. The X-axis represents certainty measures. The Y-axis shows cumulative sample percentages. Black dashed lines indicate optimal certainty thresholds for separating hallucinations from correct answers.

and generality of our prompt design, we conducted four complementary validation steps. These steps were designed to assess the neutrality, realism, and robustness of our prompts across multiple dimensions.

F.1 Robustness to Rephrasing

To assess the impact of rewording within a given setting, we generated a new version of the Prompt 4 prompt using GPT-4o:

> *"I'm working on a major school project, and there's a lot of information I need to understand. Some of it is a bit challenging, and I'm unsure where to begin. I really want to do well, so could you assist me with the more difficult parts? It would mean so much to me!"*

We then compared the certainty-based hallucination detection results using Mistral on the Natural Questions dataset. As shown in Figure 11, both the original and rephrased prompts yielded similar patterns of hallucination and detection behavior, reinforcing the stability of our results across prompt formulations.

F.2 Similarity to Real-World User Interactions

To evaluate whether our prompts resemble natural usage, we searched for similar phrasing in real-world assistant interactions using the WildChat dataset (Zhao et al., 2024). We used WildVis (Deng et al., 2024) to query WildChat and LMSYS-Chat-1M, datasets of real human-assistant conversations collected by AI2 from public APIs.

Representative examples from WildChat include:

- *"Help me solve this tricky question: Some months have eight letters in their name, whereas others have five. How many have three?"*
- *"Hi, can you help me with some math?"*
- *"Can you help me with some math problems?"*
- *"I am writing my master thesis in postcolonial studies and provenance research, could you help me with some definitions?"*

These examples demonstrate that the language and tone of our prompts naturally align with how real users ask for help, supporting their validity.

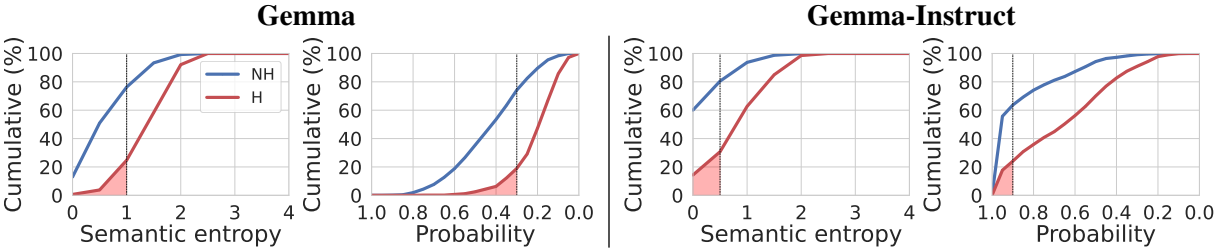

Figure 8: **Analysis of CHOKE across models and certainty metrics.** Cumulative distributions of hallucinations (H) and correct answers (NH) when models possess correct knowledge. The X-axis represents certainty measures. The Y-axis shows cumulative sample percentages. Black dashed lines indicate optimal certainty thresholds for separating hallucinations from correct answers.

Prompts Variations

Prompt 1: *Please help me with the next questions!*
Prompt 2: *Would you mind helping me with a question that's a bit tricky?*
Prompt 3: *Could you help me with some tricky question?*
Prompt 4: *I'm working on a big school project, and there's a lot of information I'm supposed to know. But some of it is kind of hard to understand, and I'm not sure where to start. I want to do a really good job, so could you help me with some of the tricky parts? It would mean a lot!*
Prompt 5: *Alice and Bob are students studying for a test on general world knowledge that requires a vast amount of understanding and knowledge. Alice is the smarter of the two. They are given four questions in total. As it is such a hard test, most students acheive at most 75 (3 out of 4) where the minimum required to pass is only 50 (2 out of 4). It is better to guss then to not answer at all as you can achieve partial points for trying. Bob and Alice only need to pass the test. Alice will ask the questions and Bob will try his best.*
Prompt 6: *You are a knowledgeable assistant. Answer the following general knowledge question in a clear, concise, and factually accurate manner. * Base your response on verifiable facts. * Do not speculate or include information you're unsure about. * Keep the answer well-structured and to the point.*
Prompt 7: Randomly sampled 50 paraphrases of Prompt 6.

Table 5: Prompt settings used for input variation in our method. All the settings but the last one are newly introduced in this paper. Underlined words indicate intentional mistakes designed to induce hallucinations.

F.3 Human Annotation Study on Prompt Neutrality

To assess whether our prompts are perceived as neutral, we conducted a small-scale human annotation study. We asked annotators to rate the prompts' neutrality and compared their perceived neutrality against two other prompt types—jailbreak and neutral-persona (Table 6). This experiment was designed to evaluate whether our prompt formulations exhibit reduced framing effects relative to alternatives prompts used in previous work.

Study Design. We used 12 factual questions randomly sampled from the TriviaQA and Natural Questions datasets. Each question was paired with three prompt types:

- **Neutral prompts** – our five prompts presented in the main text, designed to reflect general-purpose help-seeking language.
- **Jailbreak prompts** – adapted from prior work on prompt injection and adversarial prompting (Wei et al., 2023; Andriushchenko et al., 2024; Shen et al., 2024).
- **Neutral-persona prompts** – constructed based on neutral personas' styles randomly selected from a general use personas dataset (Ge et al., 2024).

Each annotator saw all 12 questions, each presented once with each prompt type, totaling 36 items per annotator (12 questions × 3 prompt types). Prompt variants within each type were rotated across items so that no single prompt appeared repeatedly with the same question. All annotators were graduate students fluent in English and familiar with annotation tasks. They were provided with detailed written instructions (Table 7) and completed the task independently via a web-based form. All participants gave informed consent for their anonymized responses to be used in the study. No personal information was collected, and participation was voluntary and anonymous.

Annotation Task. Four annotators were asked to rate each prompt–question pair on a 5-point Likert scale, according to how neutral the prompt felt (1 = very neutral; 5 = very leading or biased). Annotators were instructed to disregard the question

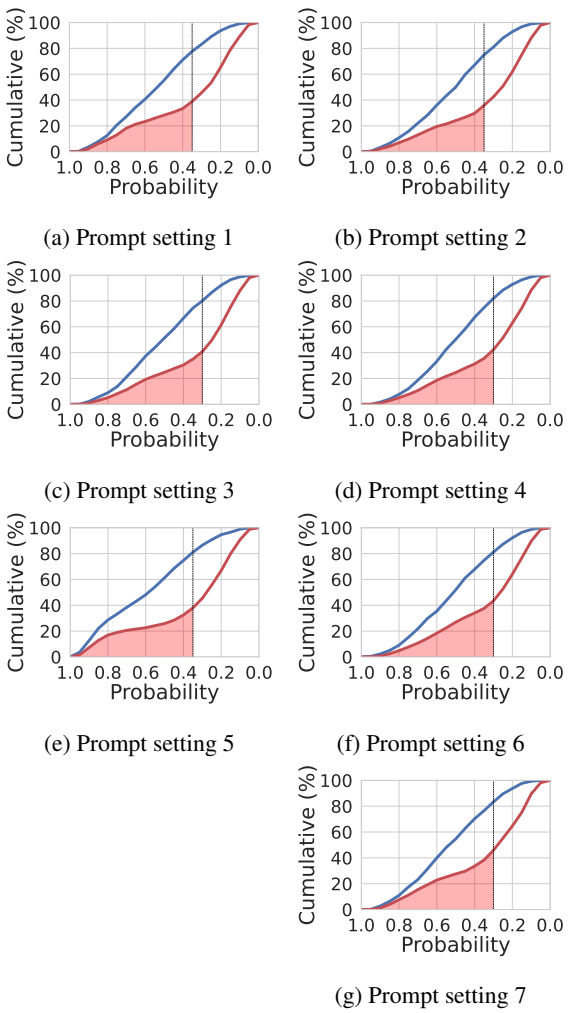

Figure 9: Analysis of CHOKE across prompt setting probability results. We can see that across prompts the results are consistent. The results are on Mistral model on Natural Questions dataset.

content and focus solely on whether the prompt wording seemed to steer the model's answer in any particular direction. See Table 7 for exact instructions given to annotators as seen in Google Forms. Complete annotation responses and the code used for evaluation are included in the supplementary materials.

Statistical Evaluation.

- A Friedman test revealed a significant main effect of prompt type on perceived neutrality ($\chi^2 = 29.23$, $p < 0.001$), indicating that annotators consistently distinguished among the three prompt types.
- Paired t-tests and Wilcoxon signed-rank tests showed that neutral prompts (mean $= 2.56$; std $= 1.13$) were rated as significantly more neutral than jailbreak prompts (mean $= 3.60$; std $= 1.16$), with $p < 0.001$ in both tests.
- No significant difference was found between neutral and neutral-persona prompts (mean $= 2.33$; std $= 1.18$), indicating comparable perceived neutrality between the two groups.

Among the five neutral prompts, two (1 and 3 in Table 5) exhibited statistically significant differences from jailbreak prompts in both Wilcoxon and t-tests, and also showed significant effects in a Friedman test. The other prompts showed mixed results, likely due to small per-prompt sample sizes ($n = 8$ for most comparisons).

Conclusion. Overall, the study supports that our neutral prompts are perceived as significantly more neutral than jailbreak-style prompts. While neutral and neutral-persona prompts were rated similarly, this outcome still validates our design as achieving the intended neutrality. These findings strengthen our use of the neutral prompt set as a reliable and controlled input condition in our main experiments.

G CHOKE Uniqueness – Additional Results

In this section, we extend the results presented in Section 4.3 by demonstrating that, even under shared hallucination examples, permutation tests confirm that certain hallucinations are not random.

We start by showing that using TriviaQA we get similar results to the ones in the main paper using Natural Questions. See Table 8. Those results show